Green Valley Estate - Database Design Phase Documentation

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1. Introduction

This document outlines the database design phase for Green Valley Estate, focusing on converting business operations (from Phase 1) into structured business rules, an Entity Relationship Diagram (ERD), and a logical data model. The goal is to create a centralized database to improve efficiency, data consistency, and security across all estate departments.

2. Business Rules

Core Operations

1. Households must pay monthly security and maintenance fees.
2. Maintenance Requests are categorized by priority (Emergency/Urgent/Routine) and assigned to staff.
3. Events require advance booking, with households receiving two free tickets quarterly.
4. Recycling is collected weekly, with participation tracked per household.
5. Safety Inspections are conducted monthly and logged with corrective actions.

Department-Specific Rules

- Security: Patrols logged with timestamps; incidents require resolution reports.
- Finance: Late fees incur 5% penalties; budgets require HOA approval.
- Environmental Management: Water usage per household is monitored monthly.

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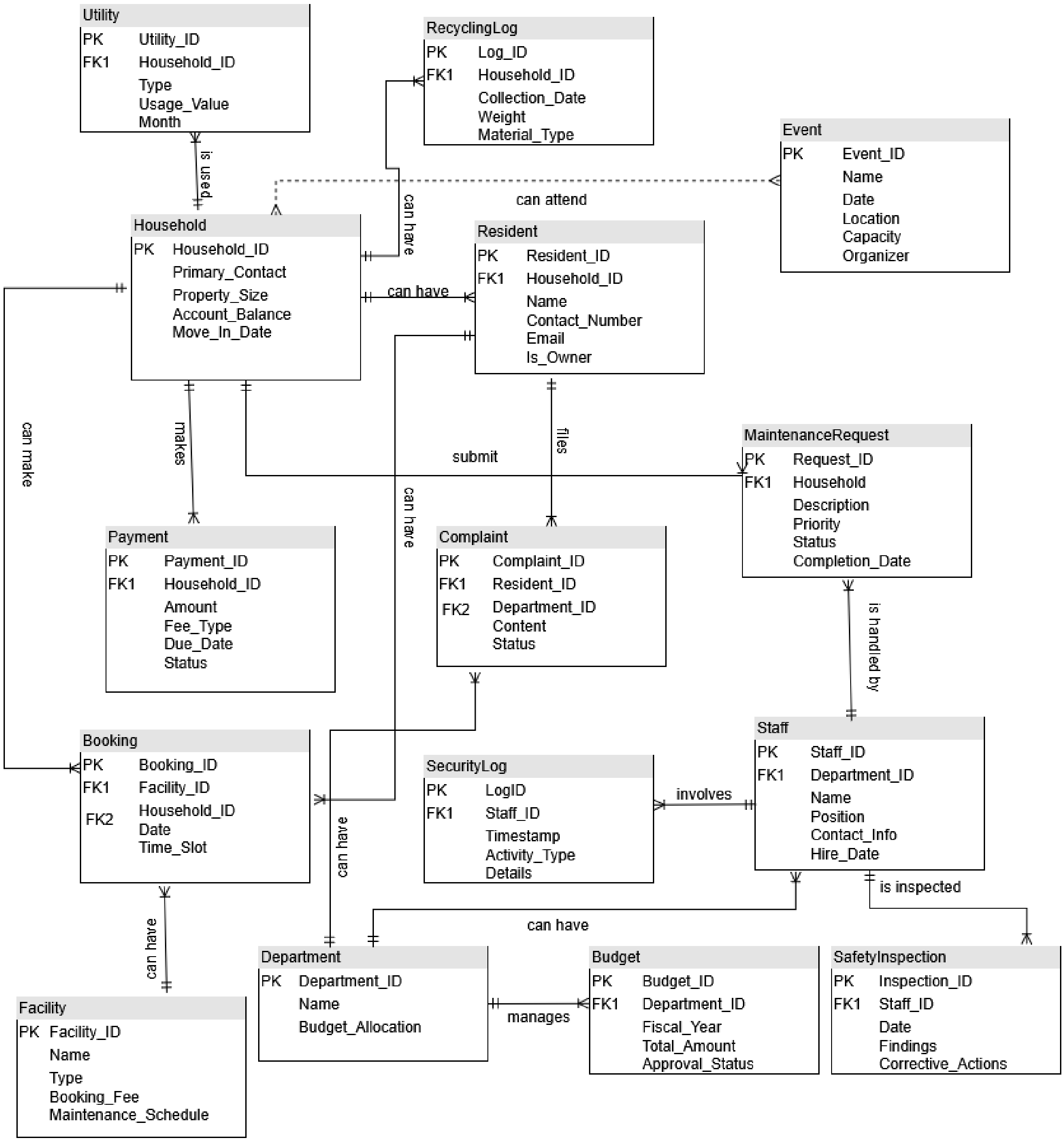
3. Entity Relationship Diagram (Crow's Foot Notation)

ERD Description: A conceptual diagram showing 15 entities and their relationships. Relationships include 1:M (Household → Payment) and M: N (Event → Registration).



4. Notes on the ER Diagram

Entities

1. Household
2. Resident
3. Payment
4. MaintenanceRequest
5. Staff
6. SecurityLog
7. Event
8. Facility
9. Booking

10. Complaint
11. Department
12. Budget
13. RecyclingLog
14. SafetyInspection
15. Utility

Key Features

- **Composite Keys**:
  - MaintenanceRequest (Request_ID + Household_ID).
- **Weak Entities**:
  - RecyclingLog (dependent on Household).
  - SecurityLog (dependent on Staff).
- **Strong Relationships**:
  - Household → Resident (1:M).
  - Staff → MaintenanceRequest (1:M).
- **Optional Relationships**:
  - Complaint → Department (a complaint may not be immediately assigned).

5. Logical Data Model

5.1. Mapping the Conceptual Model

1. Strong Entities: Household, Staff, Event, Facility.
2. Weak Entities: RecyclingLog, SecurityLog.
3. Supertype/Subtype: Not applicable (simple entity structure).
4. Composite Keys: MaintenanceRequest (Request_ID + Household_ID).

5.2. Logical Model (15 Entities)

Entity	Attributes (PK = Primary Key)
1. Household	Household_ID (PK), Primary_Contact, Property_Size, Account_Balance, Move_In_Date
2. Resident	Resident_ID (PK), Household_ID (FK), Name, Contact_Number, Email, Is_Owner
3. Payment	Payment_ID (PK), Household_ID (FK), Amount, Fee_Type, Due_Date, Status
4. MaintenanceRequest	Request_ID (PK), Household_ID (FK), Description, Priority, Status, Completion_Date
5. Staff	Staff_ID (PK), Department_ID (FK), Name, Position, Contact_Info, Hire_Date
6. SecurityLog	Log_ID (PK), Staff_ID (FK), Timestamp, Activity_Type, Details
7. Event	Event_ID (PK), Name, Date, Location, Capacity, Organizer
8. Facility	Facility_ID (PK), Name, Type, Booking_Fee, Maintenance_Schedule
9. Booking	Booking_ID (PK), Facility_ID (FK), Household_ID (FK), Date, Time_Slot

10. Complaint	Complaint_ID (PK), Resident_ID (FK), Department_ID (FK), Content, Status
11. Department	Department_ID (PK), Name, Budget_Allocation
12. Budget	Budget_ID (PK), Department_ID (FK), Fiscal_Year, Total_Amount, Approval_Status
13. RecyclingLog	Log_ID (PK), Household_ID (FK), Collection_Date, Weight, Material_Type
14. SafetyInspection	Inspection_ID (PK), Staff_ID (FK), Date, Findings, Corrective_Actions
15. Utility	Utility_ID (PK), Household_ID (FK), Type, Usage_Value, Month

Foreign Key Relationships

- Payment.Household_ID ⟶ Household.Household_ID
- MaintenanceRequest.Household_ID ⟶ Household.Household_ID
- Booking.Facility_ID ⟶ Facility.Facility_ID
- SafetyInspection.Staff_ID ⟶ Staff.Staff_ID

Conclusion

The database design for Green Valley Estate marks a critical milestone in addressing the operational inefficiencies and data management challenges identified in Phase 1. By transitioning from a fragmented manual filing system to a centralized, normalized database, the estate is poised to achieve significant improvements in efficiency, accuracy, and security across all departments.